

SOUROV HOSSEN

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Summary

Computer Science & Engineering student with a strong foundation in cybersecurity and security research. Experienced in developing security-focused projects and conducting practical security research, with strong technical and communication skills. Interested in supporting students and contributing to collaborative academic environments.

Education

Bachelor of Computer Science and Engineering (BSc in CSE) 2022 – 2026
Daffodil International University (DIU) CGPA: 3.52 / 4.00

- **Undergraduate Thesis:** A Hybrid Network Intrusion Detection System with Explainable Artificial Intelligence. Research focused on Explainable Artificial Intelligence (XAI), Network Intrusion Detection Systems (NIDS), machine learning, and deep learning-based cyber threat detection.

Higher Secondary Certificate (HSC) 2019 – 2021
Gazipur Ideal College (GIC) GPA: 4.50 / 5.00

Secondary School Certificate (SSC) 2017 – 2019
Sristy Academic School (SAS) GPA: 4.06 / 5.00

Research Interest

My research interests lie at the intersection of cybersecurity and artificial intelligence. I am committed to advancing research by integrating AI with cybersecurity to enhance the overall resilience of modern digital systems.

Research Experience

A Hybrid Network Intrusion Detection System with Explainable Artificial Intelligence *In Progress*

- Architected a hybrid XAI-driven NIDS with three detection layers for transparent cyber threat detection.
- Achieved high detection accuracy using a super-hybrid model that combines SHAP, LIME and rule-based explanations for transparent and interpretable intrusion detection.

Identity Theft and User Cybersecurity Behaviors Predicting User Cyber Risk *Submitted to Elsevier*

- Designed a SHAP-based cyber risk score framework using interpretable machine learning to predict identity theft vulnerability across users.
- Achieved high classification accuracy, finding users at risk or not with a hybrid ensemble model and identified the key factors influencing user risk profiles using SHAP-driven analysis.

Projects

DorkNio | OSINT & Google Dorking Tool GitHub Repository

- Built an advanced Google Dorking and OSINT platform that enables security researchers to generate, validate, and analyze complex search queries for authorized reconnaissance.
- Streamlined multi-source intelligence gathering through a unified interface, reducing manual OSINT effort while supporting security assessments of web assets, URLs, domains, and IP addresses.

Phishing Analyzer | ML-Based Phishing Detection & Risk Assessment System GitHub Repository

- Engineered a machine learning-powered security platform that analyzes files, emails, URLs, domains, and IP addresses to detect phishing attacks and assess threat severity.
- Automated a multi-vector phishing detection pipeline using intelligent feature extraction, classification models, and real-time risk scoring to accelerate security investigations.

ReconNio | Go-Based Reconnaissance & Security Assessment Toolkit GitHub Repository

- Developed a high-performance reconnaissance toolkit in Go featuring 19+ modules for authorized reconnaissance. Built an all-in-one security reconnaissance toolkit that consolidates multiple tools into a single platform for efficient security research.

- Designed a modular and concurrent scanning architecture with structured different types of output with console reporting to streamline security auditing and bug bounty workflows.

Achievements

Junior Cybersecurity Analyst – Cisco
CompTIA Security+ – Coursera Certificate

Network Cable – LetsDefend
eJPT – INE (In Progress)

Technical Skills

- **Fundamental Knowledge:** Data Structures, Algorithms, Database Management Systems, Object-Oriented Programming (OOP), Operating Systems, System Design and Problem Solving.
- **Programming Languages:** Python, Java, JavaScript, C, C++ and automation scripting.
- **Security & Tools:** Burp Suite, Nmap, Wireshark, Metasploit, sqlmap, Splunk, Impacket, Nikto and others.
- **Systems & Infrastructure:** Windows & Linux Administration, Active Directory, Azure, Virtual Machine.
- **Others:** OWASP Top 10, OSINT, Bash scripting, AI and ML, application development, documentation/research.